

Session 4 — Exercises: Bias & effect — can we trust it, how big is it?

Foundations of Clinical Research · Week 4 · Saturday · Neudata · #ClearDataClearImpact

This handout has six parts. You will not need a computer. For the calculations, simple division is enough; the instructor checks every answer with the 2×2 calculator.

Part	Activity	When	How
A	Spot the bias (4 scenarios)	End of section 4.1 (4 min)	Whole room / alone
B	Confounding or effect modification? (3 scenarios)	Homework, or if 4.1 runs early	Alone or pairs
C	Two 2×2 calculations	After the break (20 min)	Alone, then check with a neighbour
D	PPV in two settings	Homework, or if 4.3 runs early	Alone
E	Read a forest plot and choose the risk-of-bias tool	Pair task in 4.3 (6 min)	Pairs
F	Proposal builder 4 — exposure, outcomes and biases	End of the session (30 min)	Your group

Key words of the day: bias · confounding · effect modification (interaction) · reverse causation · risk · risk ratio (RR) · risk difference (RD) · number needed to treat (NNT) · odds ratio (OR) · sensitivity · specificity · PPV / NPV · ROC curve · forest plot · I^2 · risk-of-bias tool.

Part A — Spot the bias

Instructions. For each scenario, choose the **main** problem: **selection bias**, **recall bias**, **observer bias** or **confounding**. Then write **one practical way** to reduce it.

#	Scenario	Main bias	One way to reduce it
1	In a case-control study, mothers of babies born with a birth defect remember the medicines they took during pregnancy in more detail than mothers of healthy babies.		
2	In an unblinded trial of an early-mobilisation programme, the trial nurse who knows each patient's group decides whether the patient is "ready for discharge".		

#	Scenario	Main bias	One way to reduce it
3	A study of quality of life after ICU recruits only the survivors who come back to the follow-up clinic.		
4	In a registry, patients with higher lactate received more fluid and also died more often. The authors conclude that fluid increases mortality.		

Part B — Confounding or effect modification?

Instructions. Each study splits its result by a third factor. Compare the strata **with each other** and **with the overall (crude) result**. Decide: **confounding**, **effect modification (interaction)** or **neither**. Then say what the authors should report.

> Rule of thumb: strata similar to each other but different from the crude result → confounding. Strata different from each other → effect modification.

#	Study (hypothetical)	Overall RR	Stratum 1	Stratum 2	Your answer	What to report
1	Vasopressor X vs Y and death by day 28. Clinicians chose X mainly for patients with high lactate.	1.80	lactate < 4: RR 1.00	lactate ≥ 4: RR 1.00		
2	Early mobilisation and ICU delirium	0.70	age < 65: RR 0.50	age ≥ 65: RR 0.95		
3	Hand-hygiene programme and hospital infection	0.80	day shift: RR 0.80	night shift: RR 0.79		

Part C — Two 2×2 calculations

Instructions

1. **Draw the 2×2 table first.** Exposure (or treatment) in **rows**, with the exposed/treated group on top. Outcome in **columns**, with "outcome yes" on the left.
2. Calculate only what is asked. Round to two decimals.
3. Write **one sentence** a colleague without research training would understand.

Reminder

Measure	Formula (from the table a, b, c, d)
Risk in exposed / treated	$a / (a + b)$
Risk in unexposed / control	$c / (c + d)$
Risk ratio (RR)	risk exposed $\div$ risk control
Risk difference (RD)	risk exposed $-$ risk control
Number needed to treat (NNT)	$1 \div \text{RD}$
Odds ratio (OR)	$(a \times d) \div (b \times c)$

Calculation 1 — a randomised trial

A trial randomised 400 adults on a medical ward to a new pressure-relieving mattress (200 patients) or a standard mattress (200 patients). A new pressure injury developed in **10** patients on the new mattress and in **20** patients on the standard mattress.

	Pressure injury: YES	Pressure injury: NO	Total
New mattress	a =	b =	
Standard mattress	c =	d =	

1. Risk with the new mattress = _____ Risk with the standard mattress = _____
2. RR = _____ 3. RD = _____ 4. NNT = _____ 5. OR = _____
6. Is the OR close to the RR here? Why? _____
7. One sentence for a colleague: _____

Calculation 2 — a case-control study

Investigators selected **40** ICU patients with candidaemia (cases) and **80** ICU patients without candidaemia (controls). They looked back for parenteral nutrition. Among the cases, **24** had received parenteral nutrition and 16 had not. Among the controls, **16** had received it and 64 had not.

	Cases (candidaemia)	Controls (no candidaemia)
Parenteral nutrition: YES	a =	b =
Parenteral nutrition: NO	c =	d =

1. OR = _____
2. Why can you **not** calculate the risk of candidaemia, the RR or the NNT from this study?

3. One sentence for a colleague: _____

Note: in a case-control study the columns are the case and control groups. The OR formula $(a \times d) \div (b \times c)$ stays the same.

Part D — PPV in two settings

Instructions. The same test has **sensitivity 90%** and **specificity 80%**. Complete each table for 500 people, then calculate PPV and NPV.

> Disease present = prevalence × 500. True positives = sensitivity × disease present. True negatives = specificity × disease absent.

Setting 1 — ICU, prevalence 20%

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = $TP / (TP + FP)$ = _____ NPV = $TN / (TN + FN)$ = _____

Setting 2 — screening, prevalence 2%

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = _____ NPV = _____

1. Why is the PPV so different when the test is the same? _____
2. Calculate $LR+ = \text{sensitivity} / (1 - \text{specificity})$ = _____. Does it change between the settings?

3. Optional: repeat for a prevalence of 50%.

Part E — Read a forest plot and choose the risk-of-bias tool (pair task)

The forest plot below is the one shown in section 4.3 (six hypothetical trials of restrictive vs usual fluids; outcome death; fixed-effect pooling).

![[Forest plot of six hypothetical trials](../Resources/figures/en/sat2/forest_reading.png)]

Trial	Restrictive: died / total	Usual care: died / total
Trial 1	12 / 60	15 / 60
Trial 2	30 / 150	36 / 150
Trial 3	8 / 40	7 / 40
Trial 4	45 / 250	58 / 250
Trial 5	20 / 100	27 / 100
Trial 6	5 / 30	8 / 30

Reading the plot

1. Which trial carries the most weight? Why? _____
2. Does any single trial show a clear effect (a 95% CI that does not cross 1)?

3. Does the pooled diamond cross $RR = 1$? Write the pooled result in one plain sentence.

4. Trial 3 points the other way ($RR\ 1.14$). Should it worry you? Why or why not?

Choosing the tool. For each study, choose the risk-of-bias tool: **RoB 2**, **ROBINS-I**, **Newcastle–Ottawa (or ROBINS-E)**, **QUADAS-2** or **AMSTAR 2**.

#	Study	Tool
a	Boulet 2024 — randomised trial of restrictive vs usual fluids	
b	Hyun 2023 — registry cohort, positive vs negative fluid balance	
c	Before–after study of a central-line bundle on one unit	
d	Point-of-care lactate compared with the laboratory reference test for detecting hyperlactataemia	
e	Systematic review and meta-analysis of fluid-restriction trials	
f	Cluster-randomised trial of a hand-hygiene programme (12 wards)	

Part F — Proposal builder 4: exposure, outcomes and biases

Instructions. Complete the rest of Part 2 of your proposal (rows 1–3 were written in Session 3), then copy it into the template. For every outcome write **what**, **how** and **when**.

Our research question: _____

#	Section	What to write	Our answer
4	Exposure / intervention and comparator	What exactly is the exposure or intervention, and how is it defined or measured? Who is the comparison group?	
5	Outcomes	ONE primary outcome: what it is, how and when it is measured. 2–3 secondary outcomes.	
6	Main biases and how we reduce them	Name at least two threats (selection, information, confounding) and the step in your design that reduces each.	

Our claim at the end will be: ☐ "caused" (randomised design) ☐ "associated with" (observational design), because _____

Group self-check before you finish

- ☐ There is a comparison group, or we explain why our study is descriptive.
- ☐ There is only ONE primary outcome, with a method and a time point.
- ☐ If our design is observational, we have named the main confounders (think: confounding by indication).
- ☐ Each bias has a concrete mitigation step ("we will be careful" is not a mitigation).
- ☐ Part 2 is complete: design, setting, population (Session 3) + exposure, comparator, outcomes, biases (today).